1. Why is packet switching more suitable than circuit switching for data communication in the Internet?
2. How does multiplexing improve bandwidth utilization in computer networks?
3. Why is layered architecture important in network design?
4. How does error detection differ from error correction in data communication?
5. Why is TCP considered a reliable protocol while UDP is not?
6. How does congestion control differ from flow control in TCP?
7. Why is IP considered a best-effort delivery protocol?
8. How does ARP help in communication within a local network?
9. Why is subnetting used in IP addressing?
10. How does DNS reduce the complexity of using IP addresses for users?
11. Why is CSMA/CD not suitable for wireless networks?
12. How does encryption contribute to network security?
13. Why does increasing window size improve throughput in sliding window protocols?
14. How does NAT help in conserving IPv4 addresses?
15. Why is routing required even when direct communication links exist?
16. How does a firewall differ from an intrusion detection system (IDS)?
17. Why are sequence numbers necessary in TCP communication?
18. How does latency affect real-time applications like VoIP or video conferencing?
19. Why is HTTP considered a stateless protocol?
20. How does load balancing improve network performance and reliability?